A local proof supplement for the required Gowers inverse theorem

Current source revision, 8 November 2026. Packaging revision 3.

Target. For fixed $k \geq 3$ and $\delta > 0$, every complex one-bounded function on $[N]$ of normalized $U^{k+1}[N]$ norm at least δ correlates with a one-bounded Lipschitz test along a linear orbit on one of a fixed finite family of connected simply connected nilmanifolds of step at most k. The family and positive correlation and Lipschitz bounds are fixed before N.

Review status. The exact assembled conclusion and its current $k = 20, 39$ and proposed fixed $k = r - 1$ consumer interfaces have independent mathematical acceptance relative to the explicitly retained published inputs. Current sources incorporate the reviewed strong-family, rational-subgroup and Taylor-coordinate corrections. Historical module decisions retain their original scope. The combined source-coverage index and its Bertrand and Gaussian addenda have a separate bounded independent decision. Retained published premises remain explicit. No installation or release approval is asserted.

How to read this supplement. The assembled proof appears first. The complete supporting modules follow in the order listed below. Original page numbers restart within each module; PDF bookmarks identify them. The directory contains exact standalone TeX and PDF copies, source-use and application records, independent decisions, and an origin map. Historical author-proposal wording in a module is preserved; its later exact decision is in the review directory.

Published inputs. The seven exact published sources are registered by edition, DOI and SHA256 in SOURCE-REGISTER.json. Their private local PDF copies are separate from the proof-only archive. The unpublished GTZ erratum and draft are provenance for findings, not accepted black-box inputs. The Tointon book copy is personal-use evidence and must not be redistributed.

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